

main.py

India General Knowledge Quiz Game Source Code

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""" India General Knowledge Quiz Game A clean, interactive command-line quiz application written in Python. Demonstrates: Data Structures, Control Flow, Functions, and Type Hinting. """
from typing import List, Dict, Any
def get_quiz_data() -> List[Dict[str, Any]]:
    """Returns the list of quiz questions, options, and correct answers."""
    return [ { "question": "1. What is the national animal of India?", "options": ["A. Lion", "B. Bengal Tiger", "C. Elephant", "D. Leopard"], "answer": "B", }, { "question": "2. Which city is known as the Silicon Valley of India?", "options": ["A. Mumbai", "B. Hyderabad", "C. Bengaluru", "D. Delhi"], "answer": "C", }, { "question": "3. Which is the longest river in India?", "options": ["A. Yamuna", "B. Ganga", "C. Godavari", "D. Brahmaputra"], "answer": "B", }, ]
def run_quiz(quiz_data: List[Dict[str, Any]]) -> None:
    """Executes the quiz game loop, checks user input, and calculates score."""
    score = 0
    total_questions = len(quiz_data)
    valid_choices = {"A", "B", "C", "D"}
    print("=" * 45)
    print(" WELCOME TO THE INDIA GK QUIZ GAME ")
    print("=" * 45 + "\n")
    for item in quiz_data:
        print(item["question"])
        for option in item["options"]:
            print(f" {option}")
        # Input validation loop
        while True:
            user_answer = input("\nYour answer (A/B/C/D): ").strip().upper()
            if user_answer in valid_choices:
                break
            print("Invalid input! Please enter A, B, C, or D.")
        # Check answer
        if user_answer == item["answer"]:
            print(" Correct answer! (+1 point)\n")
            score += 1
        else:
            print(f" Incorrect! The correct answer was {item['answer']}\n")
    print("-" * 45)
    # Final Summary
    print("\n" + "=" * 45)
    print(f" QUIZ OVER! Your Final Score: {score}/{total_questions}")
    print("=" * 45)
    if __name__ == "__main__":
        questions = get_quiz_data()
        run_quiz(questions)
```